MTH 225 Exam 2 (Dr. Kim, Fall 2026)

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Rules

- The cover page should only have your name and email. Nothing else may be written on the cover page.
- No calculators, notes, books, or other outside resources.
- Eyes on your own exam.
- Scratch paper is available upon request. You may only use the provided scratch paper, and all scratch paper (used and unused) must be submitted with your exam.

Expectations

- Write with proper notation.
- Show your work in order to earn points. Show the process.

The exam length is 55 minutes. There are 100 possible points.

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(7pts) 1. Define binary relation from A to B

A binary relation from set A to set B is a subset of $A \times B$.

(7pts) 2. Define even

An integer b is even if $\exists m \in \mathbb{Z}$ s.t. $b = 2m$

(7pts) 3. Define power set

The power set of a set A is the set $P(A)$ of all subsets of A.

(7pts) 4. What do we have to do to prove $\boxed{H \subseteq K}$?

Assume $x \in H$.

Prove $x \in K$

(7pts) 5. If we know $\boxed{\text{For all } y \in S, \text{ the Sun shines on } y}$ and we know $\boxed{g \in S}$ what (if anything) can we conclude?

the Sun shines on g

(7pts) 6. What do we have to do to prove $\boxed{\text{For all } j \in V, \text{ trains stop at } j}$?

Write let $j \in V$

Then prove: trains stop at j.

For all questions on this page: Let $A = \{m^2 : m \in X\}$ and let $B = \{w \in Y \mid w^2 < 6\}$.

(7pts) (a) If we know $v \in A$, what (if anything) can we conclude?

$\exists m \in X$ s.t. $v = m^2$

(7pts) (b) If we know $n \in B$, what (if anything) can we conclude?

$n \in Y$ and $n^2 < 6$

(7pts) (c) If we know $c \in X$, what (if anything) can we conclude?

$c^2 \in A$

(7pts) (d) If we know $s \in Y$, what (if anything) can we conclude?

Nothing.

(15pts) 7. Prove: for all $a \in \mathbb{Z}$, for all $b \in \mathbb{Z}$, if $6 \mid a$ and $6 \mid b$, then $6 \mid (a+b)$.

Let $a \in \mathbb{Z}$. Let $b \in \mathbb{Z}$ Since $6 \mid a$ and $6 \mid b$. So $\exists x, y \in \mathbb{Z}$ s.t. $6x = a$ and $6y = b$. Let $z = x+y$. So $z \in \mathbb{Z}$ and $6z = 6(x+y) = 6x + 6y = a+b$. So $\exists z \in \mathbb{Z}$ s.t. $6z = a+b$. Therefore, $6 \mid (a+b)$.

(15pts) 8. Based on the definitions:

- $D = \{h \in S : h \text{ drinks tea}\}$
- $S = \{x \in B \mid x \text{ laughs}\}$
- $T = \{y \in B : y \text{ does not bowl}\}$

and using the hypotheses:

- H1: for all $y \in B$, if y skis, then y has dogs.
- H2: for all $d \in D$, d does not laugh.
- H3: for all $g \in S$, if g bowls, then g skis.
- H4: for all $m \in S$, if m drinks tea, then m does not have dogs.

prove: For all $c \in S$, if c skis, then c does not drink tea.

Let $c \in S$. Suppose c skis. Since $c \in S$, $c \in B$ and c laughs. Also since $c \in S$, by H4, if c drinks tea, then c does not have dogs. Since $c \in B$, by H1, if c skis, then c has dogs. Since c skis, c has dogs. Applying modus tollens to the unused implication, c does not drink tea. □

5:13 used